



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Student Name: RAGUL.M
Course Code: CSA0305
Course Name: DATA STRUCTURE
Course Faculty: Dr. UMA PRIYADARSINI

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Slot: A

Project Title: Improving Organ Donar-Recipient Matching Using Graph Algorithms: A Smart Organ Donation System Compared with Traditional Matching Methods

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

Review 1 Image	Review 2 Image	Module Image
		MODULE 4 ORGAN REQUEST, APPROVAL and STATUS MANAGEMENT - Request Submission: Recipients can submit organ requests by specifying the required organ, urgency level, and medical requirements. - Donor Response: Matched donors can review requests and choose to Accept or Reject the request. - Real-Time Status Tracking: The system updates requests as Pending, Accepted, Rejected, or Cancelled. - Request Coordination: Once a suitable donor accepts a request, the system updates the matching status and prevents conflicting multiple acceptances for the same recipient.
Student Contribution Module Work - Title		Demo Image
Improving Organ Donar-Recipient Matching Using Graph Algorithms: A Smart Organ Donation System Compared with Traditional Matching Methods MODULE 4 ORGAN REQUEST, APPROVAL and STATUS MANAGEMENT		
Project Description: (here you write what you did in this project (contribution) including Model Description)		
This project focuses on improving the process of matching organ donors with recipients by using graph-based algorithms instead of traditional manual or rule-based matching methods. The system models donors and recipients as nodes in a graph, with edges representing compatibility factors such as blood group, organ type, tissue matching, urgency level, and geographical proximity. Graph algorithms (such as shortest path, weighted matching, or maximum bipartite matching) are applied to identify the most compatible and optimal donor-recipient pairs efficiently. The system reduces waiting time, minimizes mismatches, and prioritizes critical cases based on urgency scores. A comparison is done between this smart graph-based approach and traditional matching methods to demonstrate improved accuracy, speed, and fairness in organ allocation. Proper validation checks ensure that only medically compatible pairs are matched, improving overall transplant success rates.		

Student Signature

Guide Signature